Advanced Coding For Data Analytics (A.Y. 23/24)

BSc in Management and Computer Science

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Project:

Data Analytics in Network Biology

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Guest Speaker:

Lorenzo Madeddu (https://www.lorenzomadeddu.com/)

Abstract

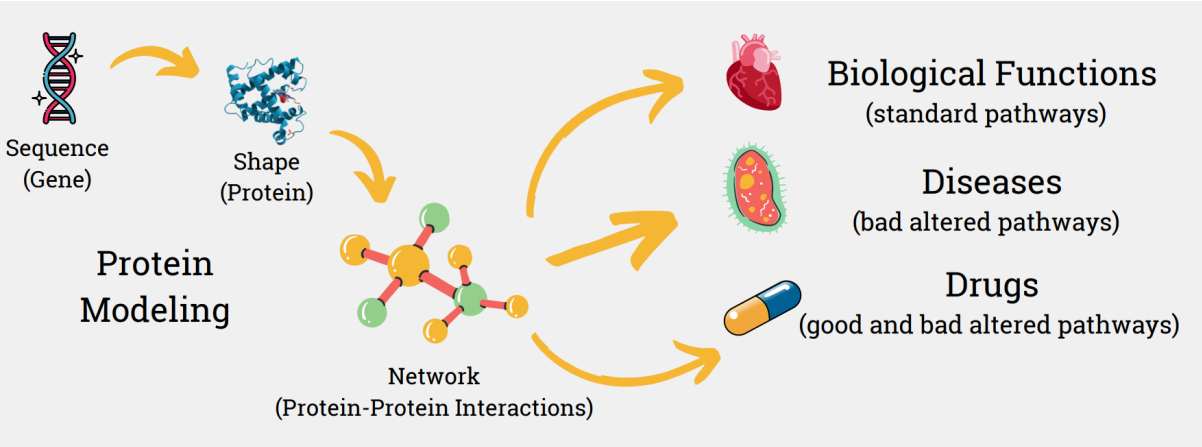
Life on earth is regulated by a complex system of interactions. Modeling those interactions through a network formalism allows researchers to discover the fundamental molecular mechanisms which drive biological processes and human diseases. In the last years, researchers have introduced novel analytic methods to cope with the complexity of this domain.

Among these researchers, we had the pleasure to host Lorenzo Madeddu, Senior data scientist (R&D) at AstraZeneca. Lorenzo developed cutting-edge solutions at the intersection of healthcare and graph mining, and they are contributing to advance the state-of-art in the bioinformatics research, with 10+ research papers.

Based on the guest speaker presentation, we propose a **group project** that provides an overview of the data and methodologies used to address biology crucial research's problems.

Introduction to Network Biology:

Many data types and many models of biological systems are best described in terms of graphs. Computational biology and bioinformatics focus on understanding the intricate network of interactions between biological entities, which is crucial for deciphering human biological processes and disease mechanisms. An example is the study of protein-protein interactions, as proteins execute various cellular functions by interacting with each other, forming complex networks.

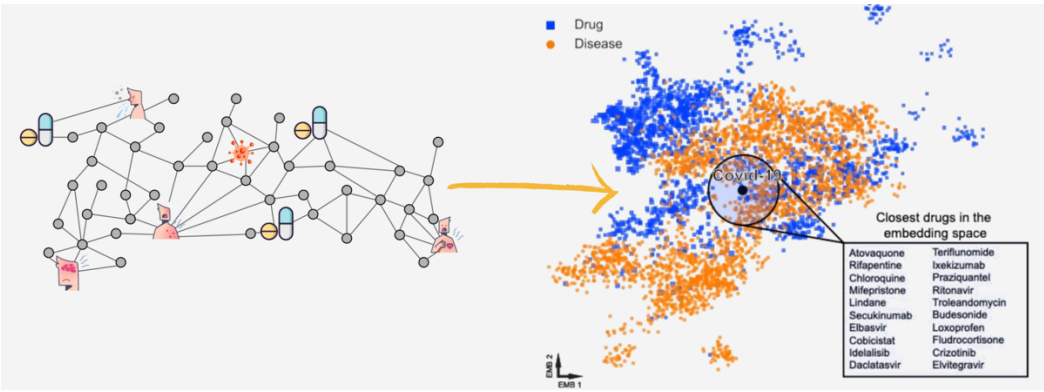


Graph modelling:

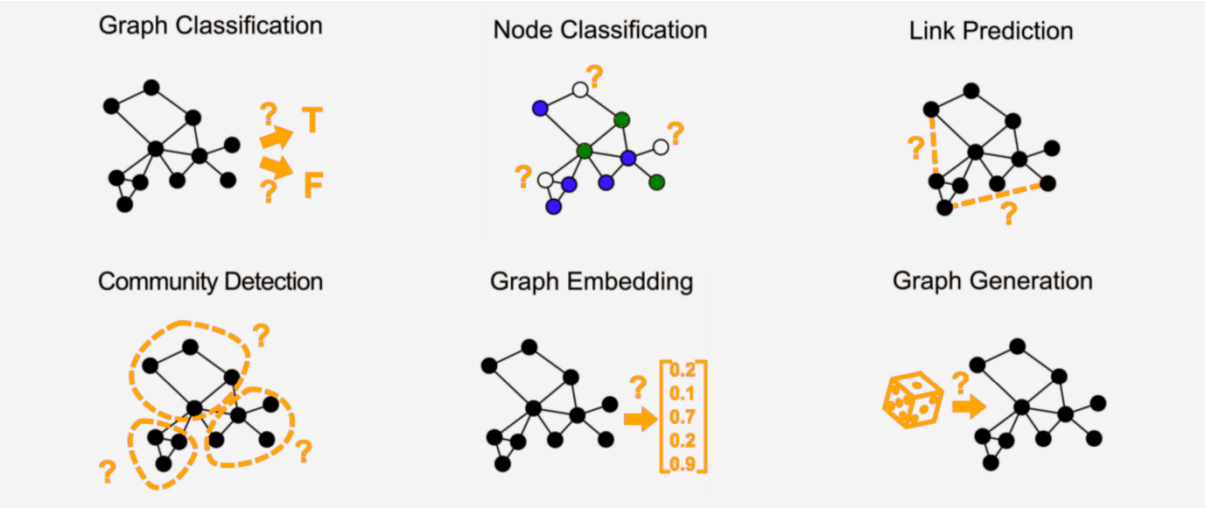
From a computer science perspective, any biological **entity** can be represented as a **node** in the graph, where their **relationships** are modelled through graph **edges**. In the example of protein-protein interaction, we can represent the network with a collection of nodes (**representing the proteins**) connected by edges (**representing their interactions**).

Graphs can heterogenous as well: nodes can represent **<u>heterogenous entities,</u>** with different types (e.g., **drug nodes** and **disease nodes)** allowing the study of their interactions (i.e., **drugs vs diseases**).

The picture below shows the example of Covid-19, where we studied which **existing drug** was more suitable for **Covid-19**:



This graph abstraction allows for the application of various (existing) graph algorithms and techniques to gain insights about our biological processes and diseases:



In this project we ask to implement some of these techniques to analyse biological graph data and produce useful insights!

Project data:

We use the publicly available **Hetionet knowledge graph** (https://het.io/about/) stored as two separate files:

1) **edges.sif**
2) **nodes.tsv**

You can download the data from here:
https://github.com/hetio/hetionet/tree/master/hetnet/tsv

The **node file (node.tsv)** lists the unique ID, name and kind for all entities in the graph.
The **edge file (edges.sif)** lists all the connections between those entities.

For example, the first entry of the edge file:

Gene::9021	GpBP	Biological Process::GO:0071357

means that the source node **Gene::9021** is connected to the target node **Biological Process::GO:0071357** via the metaedge (edge type) **GpBP**.

NOTE: *More details are provided in the Github link.*

Project goal:

For simplicity we are breaking down the project goal in three main steps, but feel free to work directly with your own schedule.

Task 1 – Constructing the biomedical knowledge graph:

Setup your Python environment and download the graph data from Github: https://github.com/hetio/hetionet/tree/master/hetnet/tsv

Remember you have two files, the **node file (node.tsv)** lists the unique ID, name and kind for all entities in the graph; and the **edge file (edges.sif)** lists all the connections between those entities.

Load the files and combine them to create your first **graph structure**. You can use any tools you prefer.
Hint: You can save the graph, so you don't need to re-build it all the times.

Task 2 – Analyzing the biomedical knowledge graph:

We now ask you to compute some metrics and analysis on the **biomedical knowledge graph**.

First of all:

1) Compute and list the total number of entities for each unique kind.
2) Compute and list the total number of each unique edge type.
3) Compute the average in-degree and out-degree, for each type of node.

The graph is actually very huge! And you noticed it contains several heterogenous entities.
Let's delve into it:

Analysis:

It's important to understand how frequent and connected a disease is, and its symptom. Let's work on that:

1) For each "**Disease**" compute the number of connections (in-degree and out-degree) and show the most **connected diseases**.
2) Compute the sub-graph of "**Disease**" and **"Symptom"**. And compute the average number of connections between a disease and a symptom.

Now you may notice some symptoms can be associated with different diseases!! Also, some diseases are very similar, let's compute the similarity of them:

3) Compute the similarity between **"type 1 diabetes mellitus"** and "**type 2 diabetes mellitus"**

4) Define and compute also the similarity between two symptoms like "**Eye Pain**" and **"Blindness"**

What "similar" means? Two diseases could be similar if they share part of their neighbours, or maybe you can do something more sophisticated ☺

Given what we learned, is now time for the final challenge:

5) Suppose you have an imaginary patient with one symptom (of your choice). Try to help them with some data analysis (e.g., study the possible diseases, select the disease, the gene, the possible compounds, the side effects, try to mitigate side effects, and so on...)

Task 3 – Visualization and reporting:

Please present all your discoveries in the presentation, using the approach and visualization methods you prefer.

Hint for a nice presentation:

- You can plot the sub-graph of all the "**diseases**" showing their connections (without the connection type) and the actual disease names. You can colour the diseases (nodes) associated with your imaginary patient.

or

- You can plot the sub-graph of your patient: the nodes will be the **symptom** connected to **possible diseases**, to **compounds** and **possible side effects**.

Project Report

The project is MANDATORY and accounts for 70% of the final grade, and we expect you to work in a group of 3 students.

Deliverable:
You must provide a presentation (in case of ppt or pdf at most 8 slides), describing your project, few implementation details, and the analysis done with the results.

You should send us (Coletta and Martino) an e-mail with a zip file containing your presentation/slides and the code, using your group id as zip name.
When you send us your project, please, use as subject "Coding2024: group X", where X is your group ID assigned by the instructors.

AI policy:
Most of the AI tools work really well and produce high quality code and software, however hallucination is around the corner (especially in state-of-art data analysis tasks). Coders should be able to understand the output of these AI tools, and use them in a conscious way.
In this course we kindly ask to code and try by yourself, made errors, and eventually learn. Copilot and chat-GPT will be much better when you can better evaluate their answers!

Note:
If you deliver the project during the last lesson of the course you will get 1 additional point. More info on luiss-learn.

References:

We refer to the slides of our guest speaker *Lorenzo Madeddu* for a more extensive introduction to data analytics and deep learning for network biology:

https://www.canva.com/design/DAF_2dY-u20/1znGq9sWzscJTE_0M9AB4A/edit?utm_content=DAF_2dY-u20&utm_campaign=designshare&utm_medium=link2&utm_source=sharebutton

A pdf version is available through the course website on luiss-learn.

We really hope you enjoyed the lecture from our guest speaker! As always, if you have any comments/questions feel free to send us an email!

Good luck!!!